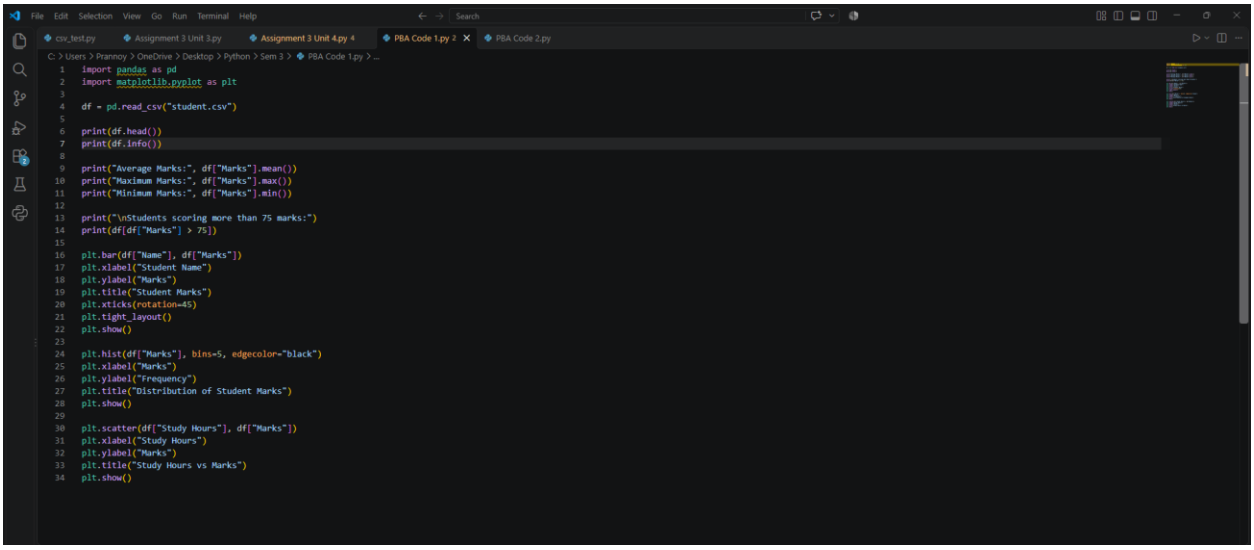
PBA 2 Code 1

Code:



```python
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("student.csv")

print(df.head())
print(df.info())

print("Average Marks:", df["Marks"].mean())
print("Maximum Marks:", df["Marks"].max())
print("Minimum Marks:", df["Marks"].min())

print("\nStudents scoring more than 75 marks:")
print(df[df["Marks"] > 75])

plt.bar(df["Name"], df["Marks"])
plt.xlabel("Student Name")
plt.ylabel("Marks")
plt.title("Student Marks")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

plt.hist(df["Marks"], bins=5, edgecolor="black")
plt.xlabel("Marks")
plt.ylabel("Frequency")
plt.title("Distribution of Student Marks")
plt.show()

plt.scatter(df["Study Hours"], df["Marks"])
plt.xlabel("Study Hours")
plt.ylabel("Marks")
plt.title("Study Hours vs Marks")
plt.show()
```

Output:

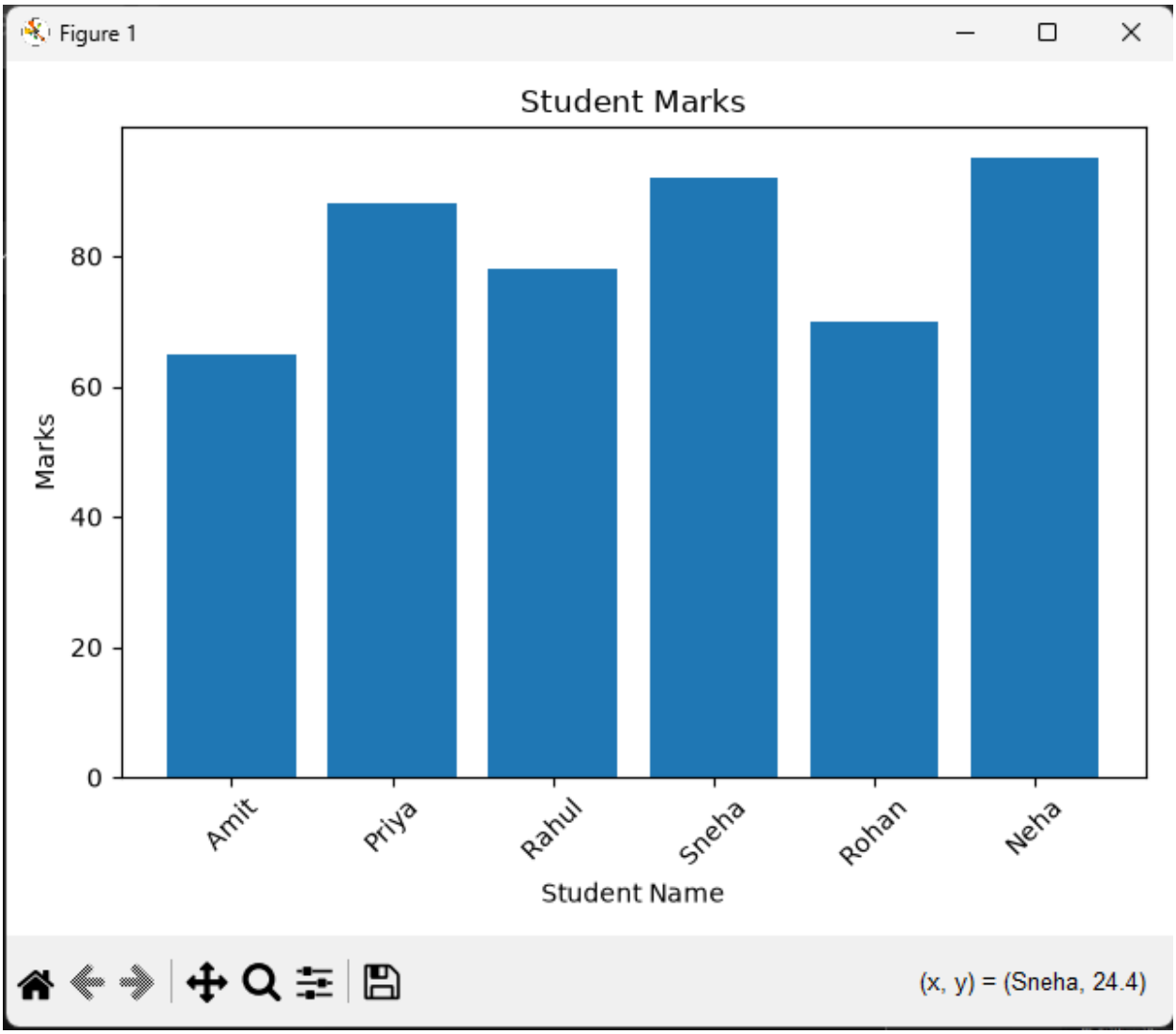


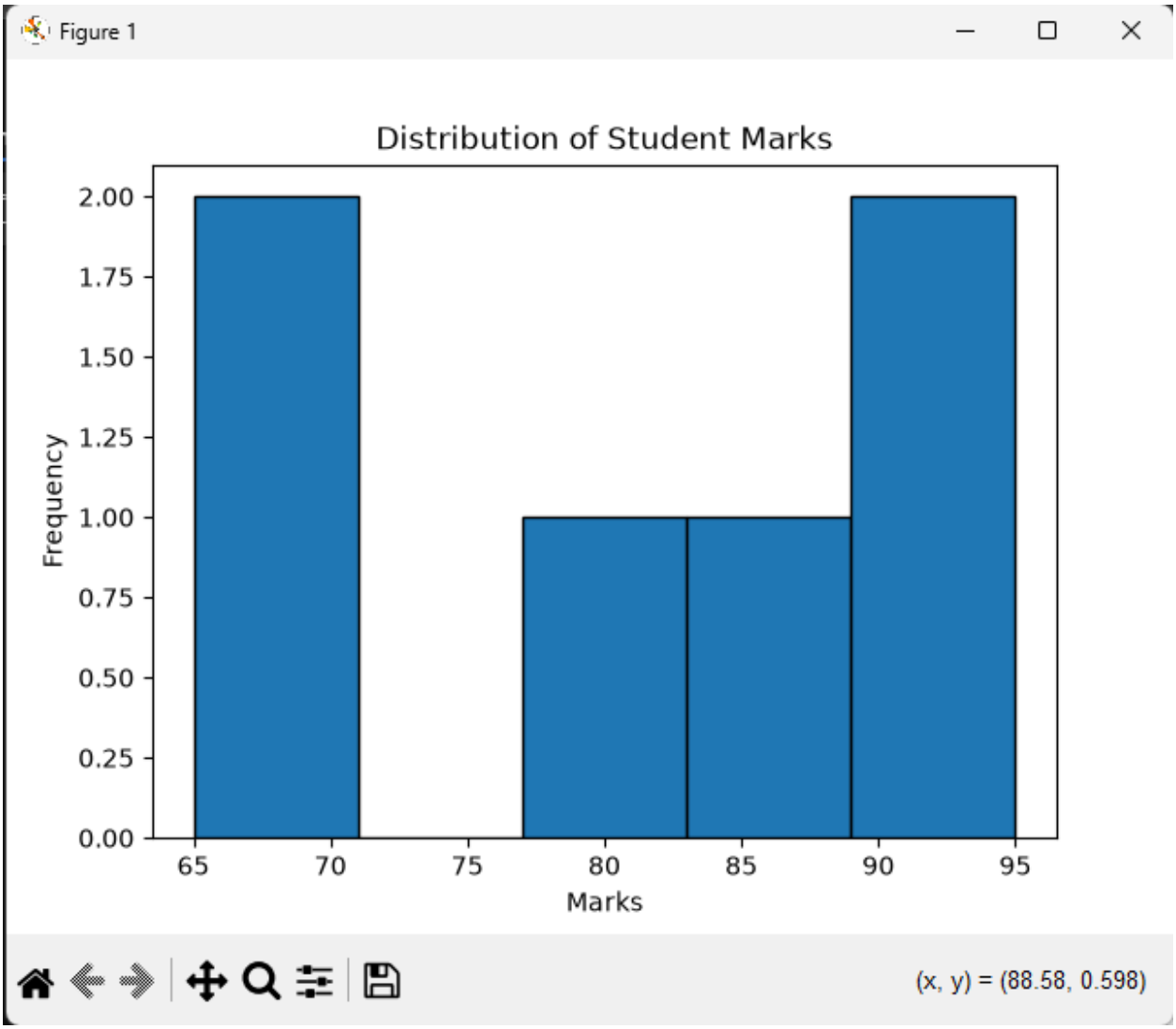
```
Problems 6  Output  Debug Console  Terminal  Ports

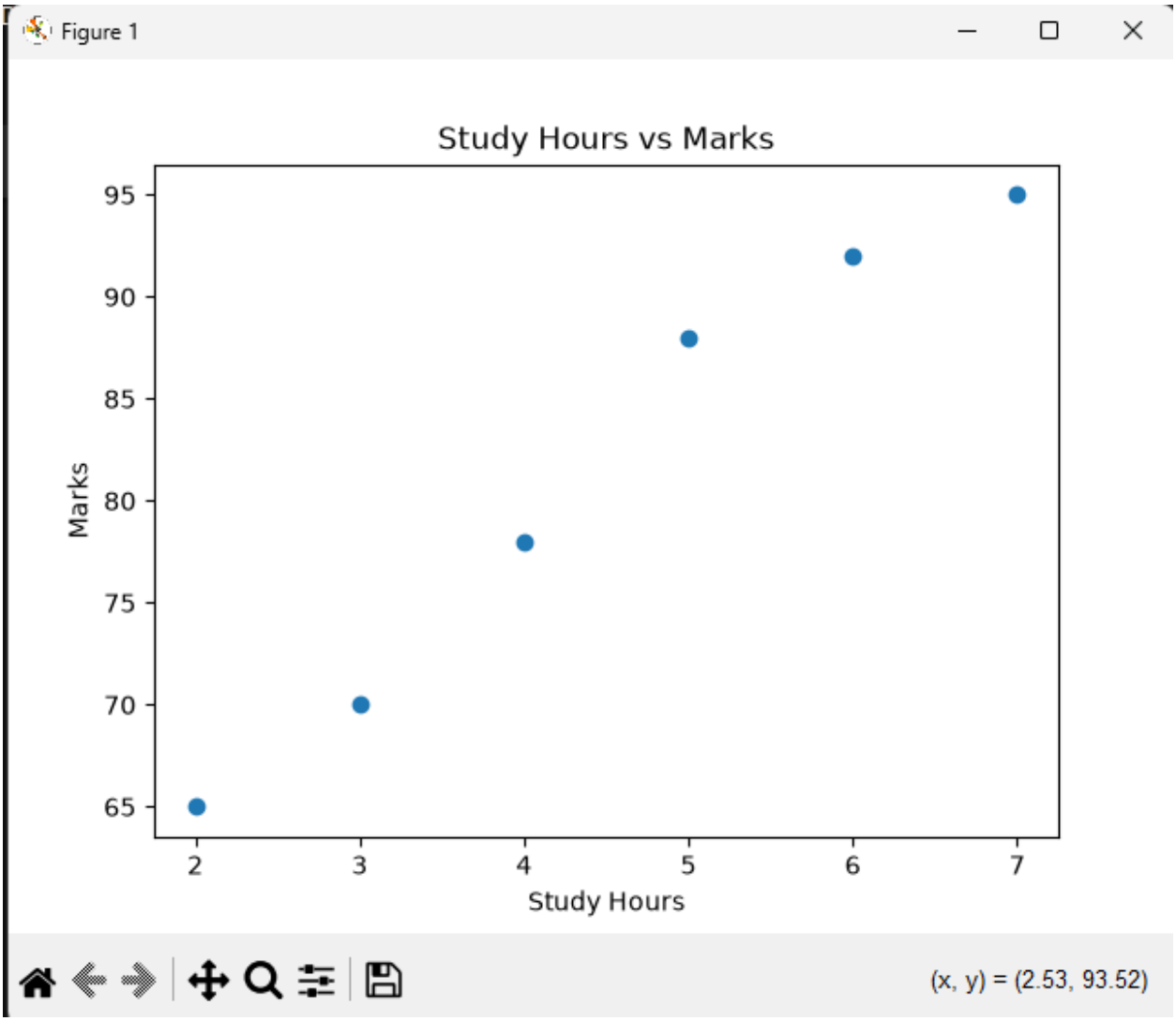
PS C:\Users\Prannoy\OneDrive\Desktop\Python\Sem 3> C:\Users\Prannoy\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/Prannoy/OneDrive/Desktop/Python/Sem 3/PBA Code 2.py"
Valid Indian Mobile Numbers:
9876543210
8765432109
7654321098
PS C:\Users\Prannoy\OneDrive\Desktop\Python\Sem 3> C:\Users\Prannoy\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/Prannoy/OneDrive/Desktop/Python/Sem 3/PBA Code 1.py"
    Name  Roll No  Study Hours  Marks
0   Amit        1            2     65
1  Priya        2            5     88
2  Rahul        3            4     78
3  Sneha        4            6     92
4  Rohan        5            3     70
<class 'pandas.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 4 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   Name         6 non-null      str
 1   Roll No      6 non-null      int64
 2   Study Hours  6 non-null      int64
 3   Marks        6 non-null      int64
dtypes: int64(3), str(1)
memory usage: 324.0 bytes
None
Average Marks: 81.33333333333333
Maximum Marks: 95
Minimum Marks: 65

Students scoring more than 75 marks:
    Name  Roll No  Study Hours  Marks
1  Priya        2            5     88
2  Rahul        3            4     78
3  Sneha        4            6     92
5   Neha        6            7     95
```

Graphs:

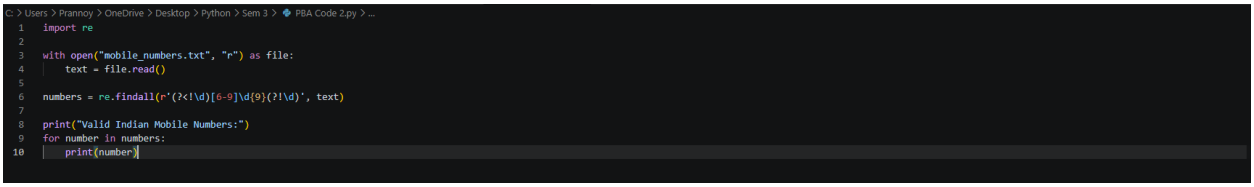

Figure 1
Student Marks
Marks
0
20
40
60
80
Amit
Priya
Rahul
Sneha
Rohan
Neha
Student Name
(x, y) = (Sneha, 24.4)


Figure 1
Distribution of Student Marks
Frequency
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
65
70
75
80
85
90
95
Marks
(x, y) = (88.58, 0.598)


Figure 1
Study Hours vs Marks
Marks
95
90
85
80
75
70
65
2
3
4
5
6
7
Study Hours
(x, y) = (2.53, 93.52)

PBA 2 Code 2

Code:



```
C: > Users > Prannoy > OneDrive > Desktop > Python > Sem 3 > PBA Code 2.py > ...
1  import re
2
3  with open("mobile_numbers.txt", "r") as file:
4      text = file.read()
5
6  numbers = re.findall(r'(?<!\d)[6-9]\d{9}(?!\d)', text)
7
8  print("Valid Indian Mobile Numbers:")
9  for number in numbers:
10     print(number)
```

Output:



```
C: > Users > Prannoy > OneDrive > Desktop > Python > Sem 3 > PBA Code 2.py > ...
1  import re
2
3  with open("mobile_numbers.txt", "r") as file:
4      text = file.read()
5
6  numbers = re.findall(r'(?<!\d)[6-9]\d{9}(?!\d)', text)
7
8  print("Valid Indian Mobile Numbers:")
9  for number in numbers:
10     print(number)

Problems 6  Output  Debug Console  Terminal  Ports

PS C:\Users\Prannoy\OneDrive\Desktop\Python\Sem 3> & C:\Users\Prannoy\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/Prannoy/OneDrive/Desktop/Python/Sem 3/PBA Code 2.py"
Valid Indian Mobile Numbers:
9876543210
8765432109
7654321098
PS C:\Users\Prannoy\OneDrive\Desktop\Python\Sem 3>
```